

task - 4

~~for task 1 and 2~~

For task-2 I have used Dijkstra algo to find the shortest path. There were N places and M roads.

So, the Time complexity is,

$$\begin{aligned} T(n) &= O(n) + O(\cancel{N} + N \log N) \\ T(n) &= O(n) + O(\cancel{N} N \log N) + O(M \log N) \\ &= O(N + M \log N) \end{aligned}$$

→ This is the Time complexity for my task-2.

Ans,